

Phase 1: Data Preprocessing Pipeline

Dataset: UNSW-NB15 | Tasks: Binary & Multiclass Classification

1. Dataset Overview

Property	Value
Numerical features	40 (+ categorical: proto, service, state)
Binary target	'label' -> 0 = Normal, 1 = Attack
Multiclass target	'attack_cat' -> 10 categories
Attack types	Normal, Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms
Key challenge	Extreme scale disparity: sbytes / dbytes reach 10 ⁹
Class imbalance (binary)	BENIGN 65,100 vs ATTACK 115,271 (ratio 1.77:1)
Class imbalance (multiclass)	Normal 65,100 vs Worms 122 (ratio 533:1)

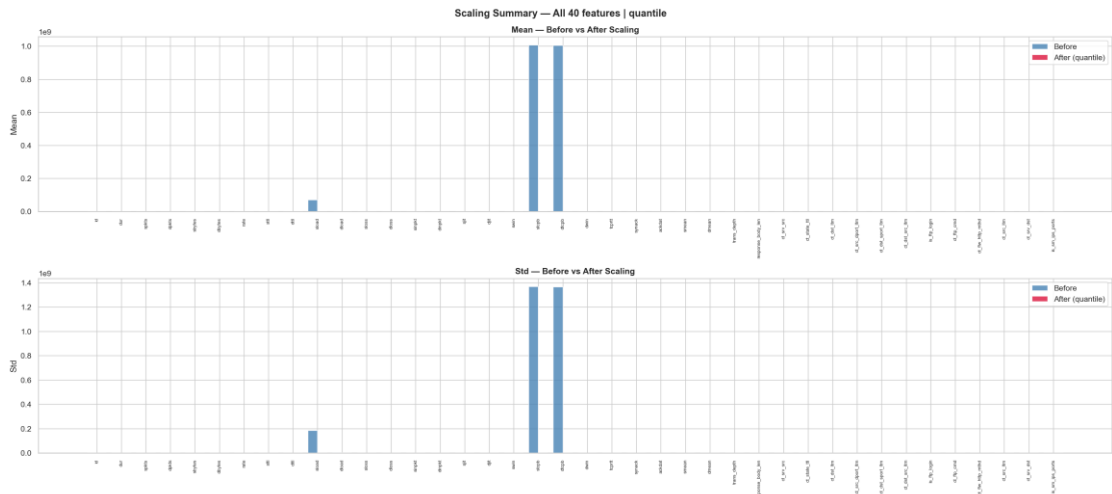
2. Pipeline Order & Design Choices

Critical rule: the scaler must be fit on the train set only, then transform applied to val/test. Oversampling is applied to the train set only, after the split. Any deviation causes data leakage. This is your recommendation!

Step	Operation	Choice	Justification
1	Missing values	drop / mean	drop preserves statistical integrity; mean retains all rows
2	Categorical enc.	OneHot / Target	OneHot for low cardinality; Target encoding fit on train only
3	Train/Val/Test split	Stratified 70/10/20	Preserves class proportions critical for 533:1 imbalance
4	Scaling	QuantileTransformer	Non-parametric; robust to log-normal distributions and 10 ⁹ outliers
5	Oversampling	SMOTE / SMOTETomek	SMOTE for binary (×1.77); SMOTETomek or ADASYN for multiclass (×533)

3. Scaling Results on UNSW-NB15

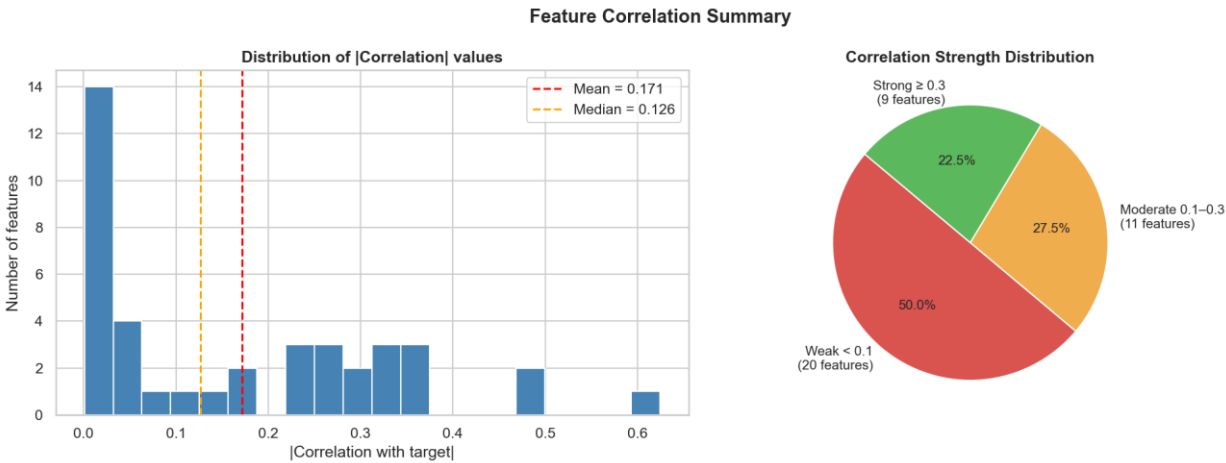
Before scaling, sbytes and dbytes dominate the feature space with means around 10⁹, rendering all other features invisible at the same scale. The QuantileTransformer maps every feature to a uniform [0, 1] distribution, effectively eliminating magnitude dominance and ensuring stable gradient flow through the both task (binary and multiclass).



Scaler	Handles outliers	Fixes skew	Recommended for IDS
QuantileTransformer	Yes	Yes	Yes, optimal for network traffic features

4. Feature Correlation Analysis

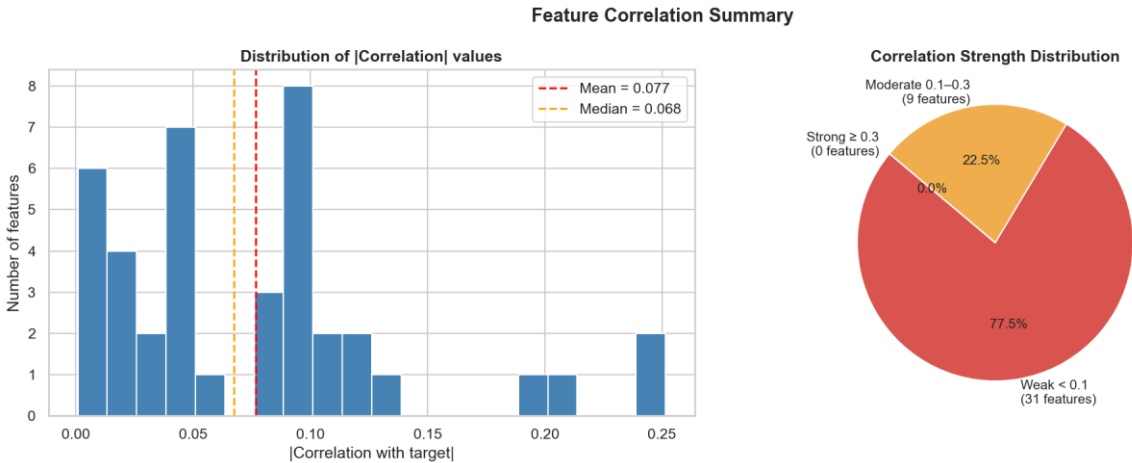
Correlation analysis serves two purposes: identifying informative features (feature ↔ target) and detecting redundant ones (feature ↔ feature). Results on UNSW-NB15 (40 features, Pearson):



Category	Threshold	Count	Action
Strong correlation with target	≥ 0.3	9	Keep highly informative
Moderate correlation	0.1–0.3	11	Keep useful signal
Weak correlation	< 0.1	20	Candidate for removal
Inter-feature correlation (redundant)	> 0.95	9	Drop one of each pair

Redundant features: sbytes, dbytes, sloss, dloss, dwin, ct_src_dport_ltm, ct_dst_src_ltm, ct_ftp_cmd, ct_srv_dst. Notably, sbytes and dbytes are both dominant in magnitude AND redundant with each other a double reason to handle them carefully.

Limitation: Pearson correlation only captures linear relationships. Complement with Mutual Information (mutual_info_classif) to detect non-linear dependencies before any final feature selection.

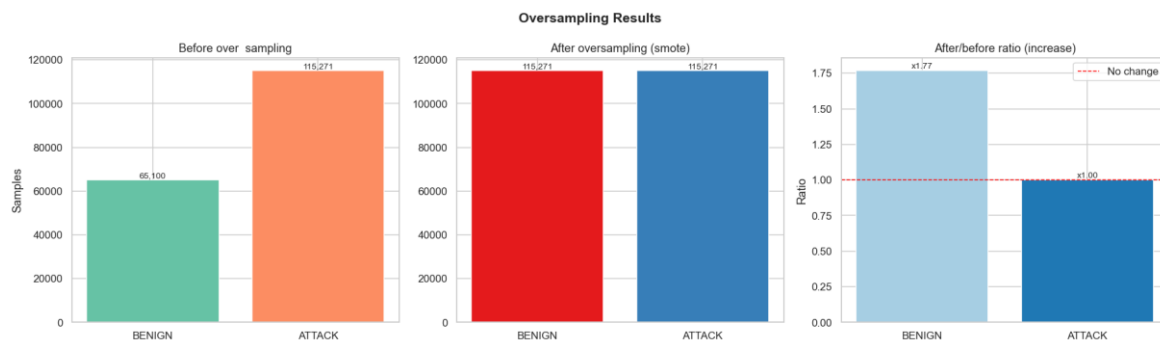


In the multiclass setting, Pearson correlation with the encoded target yields a mean of 0.077 with no strongly correlated features (≥ 0.3), compared to 9 in the binary case. This degradation is expected: Pearson assumes a linear relationship with a continuous target, whereas multiclass labels are nominal integers with no ordinal meaning. Mutual Information is the appropriate metric for this configuration.

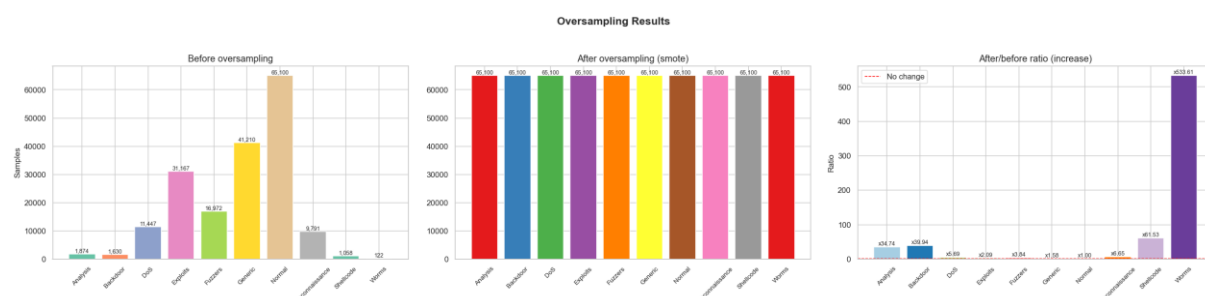
5. Oversampling Results

Binary task (ratio 1.77:1)

Class	Before SMOTE	After SMOTE	Ratio
BENIGN	65,100	115,271	$\times 1.77$
ATTACK	115,271	115,271	$\times 1.00$ (unchanged)
Total	180,371	230,542	$\times 1.28$



Multiclass task (ratio)



Class	Before SMOTE	After SMOTE	Increase Ratio
Analysis	1,874	65,100	$\times 34.74$
Backdoor	1,630	65,100	$\times 39.94$
DoS	11,447	65,100	$\times 5.69$
Exploits	31,167	65,100	$\times 2.09$
Fuzzers	16,972	65,100	$\times 3.84$

Generic	41,210	65,100	×1.58
Normal	65,100	65,100	×1.00
Reconnaissance	9,791	65,100	×6.65
Shellcode	1,058	65,100	×61.53
Worms	122	65,100	×533.61
Total	302,249	651	